

Repomodore - Software Requirements Specification

Version 1.0 | 29 September 2026 | MVP

1. Purpose and system boundary

This specification defines the mobile-first, cross-platform MVP for Repomodore on iOS, Android, and web. Mobile is the design and release priority. Each client timer and its local data work without an account. An optional Clerk identity and server API provide cross-device sync using Turso. Only authenticated server code can access Turso credentials.

2. Actors and assumptions

- Guest: uses both modes with local storage, export, and local deletion.
- Signed-in local user: authenticates with Clerk but has not enabled sync.
- Sync user: explicitly enables remote activity and settings sync.
- Platform notifications require permission and may be delayed or unavailable; the in-app state remains authoritative.
- Timing is based on wall-clock deadlines, not the count of setInterval callbacks.

3. Functional requirements

ID	Requirement	Acceptance condition
FR-01	Offer Study and Study + Squats from the timer screen.	Guest can select and run either mode without authentication.
FR-02	Configure focus 1-180 min; break 1-60 min; rep target 1-100. Defaults: 45, 5, 10.	Invalid entries are rejected; changed values apply to the next phase.
FR-03	Provide Start, Pause, Resume, Reset.	Pause freezes remaining duration; Reset asks before discarding progress.
FR-04	Study mode completes after its focus deadline and records one focus block.	No movement prompt or break is created.
FR-05	Study + Squats transitions from completed focus into a timed movement break.	5-minute break starts and displays goal 10 by default.
FR-06	Allow Done once or Skip during movement break, with an Other movement or Rest only choice.	Done records goal reps once; Skip/Rest only records zero; countdown remains active.
FR-07	Finish a break at its deadline or via End break; prepare next focus.	Next focus starts only on explicit Start.
FR-08	Show daily and 7-day totals of completed focus blocks, focused minutes, reported reps and skipped breaks.	Totals derive from records and reflect timezone-local calendar days.

ID	Requirement	Acceptance condition
FR-09	Persist timer state, settings, and activity locally on each platform.	App restart or web reload restores correct phase/remaining time, including elapsed time while away.
FR-10	Permit optional Clerk login and an independent opt-in sync toggle.	Login alone does not upload history.
FR-11	On opt-in, upload existing local records and reconcile remote records by stable ID.	Second device shows the same completed history without duplicates.
FR-12	Show pending/failed/synced state and retry after connectivity returns.	Failed writes remain queued locally; user sees status.
FR-13	Provide JSON export and separate local-data and remote-data deletion actions.	Export is readable; destructive actions require confirmation and show outcome.
FR-14	Provide optional sound and platform notification controls.	Denied permission does not affect timer progression.
FR-15	Support the same essential flow on iOS, Android, and web.	Each platform completes both timer modes and local-only history; sync interchanges records across platforms.

4. Timer state machine

States: `idle_focus`, `running_focus`, `paused_focus`, `running_break`, `paused_break`, `ready_focus`. In Study mode, focus completion goes to `ready_focus` and records a block. In Study + Squats, it records the block and goes to `running_break`. Break completion goes to `ready_focus`. A break response is a separate field: `pending`, `done`, `skipped`, or `rest_only`. If no response at break end, record skipped. Reload recomputes elapsed time from deadline; if multiple phases would have elapsed while the app was closed, complete only the current timed phase and land in `ready_focus` (never fabricate multiple sessions).

Each activity record has a client-generated UUID, `mode`, `started_at`, `completed_at`, `focus_seconds`, `break_seconds` where relevant, `movement_label`, `rep_goal`, `reported_reps`, `break_outcome`, `updated_at`, and `deleted_at` nullable. Store timestamps as UTC ISO-8601 and derive day buckets in the viewer's local timezone. Settings and active timer are separate records. Only completed focus phases enter totals. Reset during a break does not erase the completed focus record.

5. Sync and privacy requirements

The server verifies each Clerk session and scopes every Turso query to the authenticated Clerk user ID. Clients never hold a Turso auth token. Records use UUID upsert for idempotency; deletion uses tombstones to avoid reappearing across devices. For concurrent edits, choose latest `updated_at` for settings and immutable activity events; resolve deletion over a prior event. The server assigns or validates monotonic update timestamps to reduce clock-skew conflicts. Initial sync presents an opt-in explanation, uploads local data, then downloads remote data and merges by UUID. A fresh device requests remote data only after sync is enabled. Sync off stops requests and keeps local records. Account deletion removes remote records and user settings; it does not silently delete local data. No personal health or biometric data is required. Native apps store records in device-local persistent storage; web stores them in IndexedDB. All share the same versioned record schema.

6. Nonfunctional requirements

ID	Requirement	Target
NFR-01	Accessibility	VoiceOver, TalkBack, and keyboard operation; labeled controls, visible focus, contrast at least WCAG AA; avoid announcing every timer second.
NFR-02	Responsiveness	Design for 320px mobile width first, adapt to tablet and desktop; thumb-friendly primary actions and touch targets at least 44px.
NFR-03	Timer correctness	Deadline-based countdown with at most 1 second display error while foregrounded; restore elapsed time on return.
NFR-04	Offline behavior	Installed mobile apps and previously loaded web app continue locally; sync queues until connectivity returns.
NFR-05	Security	HTTPS, server-side Clerk validation, user-scoped queries, input validation, no secret in client bundle.
NFR-06	Privacy	Default local-only; explicit opt-in; export and deletion controls; minimum collected fields.
NFR-07	Reliability	Idempotent sync retry; no duplicate totals; retain unsynced records through failure.

7. Verification scenarios

1. Run Study for a shortened test duration; complete it and verify one block, no break.
2. Run Study + Squats; finish focus; tap Done twice; verify one goal-sized rep record and uninterrupted break.
3. Skip a break, let the timer expire, and verify zero reps and ready_focus.
4. Pause, restart the app or reload web, resume; background the app; verify deadline-based remaining time.
5. Use local-only anonymously, sign in without opt-in, and verify no remote activity upload.
6. Opt in with prior local records; sync on a second device; verify matching totals and UUID deduplication.
7. Go offline, complete a block, reconnect, and verify eventual sync without losing local data.
8. Turn sync off, clear remote history, and verify local data stays until separately deleted.
9. Run both modes on iOS, Android, and web at narrow/mobile viewport; verify accessible controls and correct cross-platform sync.

8. Traceability

Study-only flow: FR-01/03/04/08/09. Combined loop: FR-01/02/03/05/06/07/08. Data choice: FR-09/10/11/12/13 and NFR-04/05/06/07. Cross-platform accessibility: FR-14/15 and NFR-01/02/03.